

The Number Barrel: An Addition and Subtraction Toy

MECE3408 Computer Graphics and Design

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I. Background Research & Market Analysis

The current market for early childhood numeracy tools is dominated by two extremes: low-tech manipulatives (like abaci or flashcards) and high-tech digital applications.

- **The Gap:** Digital tools often lack the tactile feedback necessary for young, kinesthetic learners and incur concerns regarding unsupervised screen time. Conversely, physical toys often consist of numerous loose parts that are easily lost or lack an automated "feedback" mechanism to show how numbers relate to one another, specifically in the context of subtraction.
- **The Solution:** The **Number Barrel** bridges this gap by utilizing a simplified version of the Leibniz calculator mechanism. Unlike existing mechanical toys, the Number Barrel is specifically "capped" for the developmental needs of a six-year-old (operating within the 0–20 range). This prevents cognitive overload and eliminates the "negative number problem" common in uncapped mechanical calculators, where a user can accidentally crank backward past zero into a non-intuitive sequence. Material selection for this toy was based on balancing durability, safety and portability. This led us to select a majority low density polyethylene (LDPE) assembly, which would have the benefit of lesser mass when compared with HDPE, but still retain sufficient strength and finish requirements.

II. Design Operation

The Number Barrel operates on a series of intermeshing gear trains designed to translate rotational motion into digit increments.

1. **Selection & Input:** The user slides a "selector" gear along a shaft to choose a digit (1–9). By rotating the main drum wheel one full counterclockwise rotation, the selection gear engages with the counting gears.
2. **The Addition/Subtraction Logic:** A 1:1 gear ratio ensures that one rotation of the drum corresponds to the selected numerical value. Adding is achieved through counterclockwise rotation, while subtraction is mapped to the clockwise direction of the second term, providing a tactile association with "forward" and "backward" operations.
3. **The Carry Mechanism:** To reinforce the base-10 system, the "units" gear features a carry pin. Upon completing a full 360 degree rotation (representing 10 units), the pin meshes with the "tens" gear, incrementing it by one.
4. **The Geneva Stopwork:** To maintain the 0–20 range, a Geneva drive mechanism acts as a mechanical limit. The stopwork gear features a specific geometry (concave and convex portions) that physically blocks further rotation once the limit is reached. This is also the mechanism that prevents any negative numbers that could be caused if the user were to crank the barrel clockwise to a number larger than the first selected term.

III. Design Intent and Modeling

The primary design intent was **educational autonomy**. The device was modeled to be entirely self-contained, eliminating loose parts that could be lost during transport.

- **Tactile Mapping:** In modeling the device, we prioritized the "feel" of the math. The selection gear and the visual movement of the number wheels were designed to be visible through the assembly, allowing the child to witness the mechanical logic of addition.
- **Safety and Durability:** The outer "barrel" casing was modeled to be robust and ergonomic for small hands, ensuring that the internal precision of the gears is protected from external impact. Moreover, the resistance of the assembly was considered at every step, considering where parts might be sensitive to

breaking with extreme force, or where parts might get dislodged. Gears on the external part of the toy were kept smooth and large to avoid potential accidents with small fingers.

IV. Engineering Challenges & Decision Making

During the modeling and prototyping phase, several mechanical challenges required iterative design changes.

Gear Stability and Displacement

A significant difficulty encountered was the stability of the gears on their respective shafts. Initial iterations allowed for horizontal shifting; if the toy was flipped or shaken during transportation, the gears would come out of alignment, causing the mechanism to jam. To resolve this, we implemented securing ribs within the internal housing.

- **Structural Modification:** We opted to split the shafts into multiple components. This allowed us to slip the gear onto the central diameter and then screw the shaft segments together. This "sandwich" approach effectively locked the gears in place axially while allowing them to rotate freely, ensuring the toy remains functional even after rigorous handling.

Overengineering Concerns

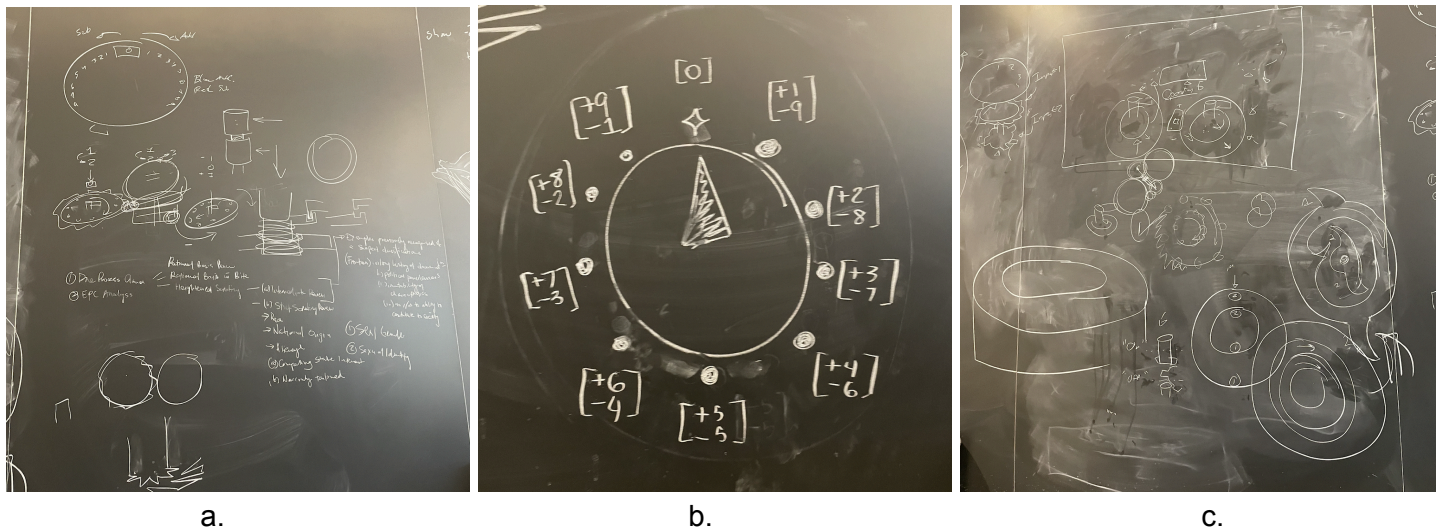
An issue that we took into consideration as well was trying to keep the design as simple as possible, so as to be manufacturable, rather than overengineering. Even with our final design, we recognize that some parts, such as our shaft solution, might have been more easily solved with a standard part, such as a c-clip, rather than splitting the shaft into multiple parts. However, our reasoning remains sound, as we struggled to find perfectly appropriate standard parts on sites like McMaster. While, in retrospect, some parts could have been simplified further, the function and intent of the ultimate design was met.

In fact, having so many component parts made it very difficult to do a motion study, as it became a lot to process.

v. Concept Drafting

Before exploring the mechanics of the Leibniz calculator, we intended to design our own mechanical calculator with two term inputs and directional operation selection. Figure 1a. shows the initial internal sub-assembly, with some ideas to solve issues such as backdriving and summing/subtracting selection. Figure 1b. shows an example of the secondary term selection wheel, and Figure 1c. Shows examples of the external assembly, as well as more attempts to solve issues with the order of operations when subtracting.

Figure 1. Initial Drafting Concepts of Mechanical Calculator



After research into the Leibniz calculator, we refined the mechanical calculator idea into what became the Number Barrel. We adapted our designs to use a single input term mechanism, and directional operation selection. Then, rather than having one large wheel with all possible outputs, we used the “carrying pin” from the Leibniz design to have “ones” and “tens” output gears. Figure 2 shows the updated original concept drawn on a blackboard during discussion.

Figure 2. Drafting Concept for the Number Barrel

